

Turán-flavored stuff

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All graphs here are simple.

1 Mantel's theorem and Turán's theorem

Our goal throughout this lecture will be to think about the following question:

Problem 1.1

For a positive integer n and graph H , let $\text{ex}(n, H)$ be the largest possible number of edges in an n -vertex graph that does not contain H as a subgraph (such graphs are called H -free). How does $\text{ex}(n, H)$ grow as $n \rightarrow \infty$?

To start with, let's consider a very simple choice of H :

Problem 1.2

What is $\text{ex}(n, K_3)$, the largest possible number of edges in an n -vertex triangle-free graph?

In the early 1900s, Willem Mantel proved Mantel's theorem, namely that

$$\text{ex}(n, K_3) = \left\lfloor \frac{n^2}{4} \right\rfloor.$$

We shall present two proofs that $\lfloor n^2/4 \rfloor$ is an upper bound on the number of edges.

Proof 1. Let G be a triangle-free graph with m edges. Let xy be an edge of G with endpoints x and y . Then, the set $N(x)$ (the vertices connected to x) is disjoint from $N(y)$, meaning

$$\deg x + \deg y \leq n$$

for all $x, y \in V$. Summing over all edges in G gives

$$\sum_{xy \in E} \deg x + \deg y \leq mn.$$

Since each vertex x appears in $\deg x$ edges, the summation equals

$$\sum_{x \in V} (\deg x)^2.$$

Applying the Cauchy-Schwarz inequality then gives

$$mn \geq \sum_{x \in V} (\deg x)^2 \geq \frac{1}{n} \left(\sum_{x \in V} \deg x \right)^2 = \frac{(2m)^2}{n},$$

which simplifies to $m \leq n^2/4$, but we can take the floor on the right because m is an integer. $\square$

Proof 2. Let v be a vertex of maximal degree, let $A = N(v)$, and let B be the vertices that are not in A . Then, there are no edges connecting two vertices of A (else we would have a triangle), meaning every single edge in G has an endpoint in B . Thus

$$m \leq \sum_{x \in B} \deg x \leq |B| \cdot \max_{x \in B} \deg x = |B| \deg v = |A| \cdot |B|,$$

but since we know that $|A| + |B| = n$ is fixed, we can apply the AM-GM inequality to get $m \leq |A| \cdot |B| \leq n^2/4$, at which point taking floors finishes. $\square$

To show that this bound on $\text{ex}(n, K_3)$ is sharp, we need to find a graph where equality holds. To do this, let's trace through the proofs to see what properties a maximal graph would need to preserve equality the whole way.

In the first proof, we would need $\deg x + \deg y = n$ for all edges for the mn upper bound to be sharp. This happens iff G is a complete bipartite graph on n vertices. From here, it's easy to check that the complete bipartite graph that maximizes the number of edges is $K_{\lfloor n/2 \rfloor, \lceil n/2 \rceil}$.

Similarly, in the second proof,

$$m = \sum_{x \in B} \deg x$$

can only hold if G is bipartite, and then the upper bound on this sum requires the graph to be complete bipartite, from which we recover the same unique maximal graph.

Exercise 1.1. Go do some of the problems in section 1.1 of the book.

One next step is to look at arbitrary cliques K_{r+1} .

Problem 1.3

What is $\text{ex}(n, K_{r+1})$?

Analogous to Mantel's theorem, we can immediately see that the n -vertex r -partite graph with part sizes differing by at most 1, which we will call the **Turán graph** $T_{n,r}$, is K_{r+1} -free. It turns out that this is again the best we can do:

Theorem 1.4 (Turán's theorem). The Turán graph is the unique maximal K_{r+1} -free graph, and

$$\text{ex}(n, K_{r+1}) \leq \left(1 - \frac{1}{r}\right) \frac{n^2}{2}.$$

Moreover, this bound is asymptotically tight for fixed r as $n \rightarrow \infty$:

$$\text{ex}(n, K_{r+1}) = \left(1 - \frac{1}{r} - o(1)\right) \frac{n^2}{2}.$$

Proof 1. This proof mirrors the first proof of Mantel's theorem. We will induct on r , with $r = 0$ being trivial.

To start with, let v again be a vertex of maximal degree in a K_{r+1} -free graph G , and again let $A = N(v)$ and $B = V \setminus A$. Then, A is K_r -free (else the K_r along with B makes a K_{r+1}), so $e(A) \leq e(T_{|A|,r})$. On the other hand, the number of edges that have an endpoint in B can again be bounded by

$$\sum_{x \in B} \deg x \leq |B| \cdot \max_{x \in B} \deg x = |A| \cdot |B|,$$

implying that equality happens when G is complete r -partite.

To show that the Turán graph is the specific r -partite graph that maximizes the number of edges, we can just observe that if two parts of the graph have counts of vertices that differ by more than 1, then moving a vertex from the more populous part into the less populous one will increase the number of edges. $\square$